Squat - Linear Regression Model - Peak Power

Model Summary - Peak Power T2

Model	R	R^2	Adjusted R^2	RMSE	Durbin-Watson		
					Autocorrelation	Statistic	p
M_0	0.000	0.000	0.000	112.14	−0.364	2.489	.310
M_1	0.719	0.517	0.483	80.66	−0.052	2.073	.918

ANOVA

Model		Sum of Squares	df	Mean Square	F	p
M_1	Regression	97534	1	97534	14.99	.002
	Residual	91093	14	6507		
	Total	188627	15			

Note. The intercept model is omitted, as no meaningful information can be shown.

Coefficients

Model		Unstandardized	Standard Error	Standardized	t	p	95% CI		Collinearity Statistics	
							Lower	Upper	Tolerance	VIF
M_0	(Intercept)	134.375	28.035		4.793	< .001	74.620	194.129		
M_1	(Intercept)	363.235	62.457		5.816	< .001	229.279	497.191		
	Abs Delta Peaks-Time Env % Cycle_T7	−4.089	1.056	−0.719	−3.872	.002	−6.354	−1.824	1.000	1.000

Note. The following covariates were considered but not included: Abs Delta Onset (s)_T7, Motion EMD Ag %Cycle_T7, Co-contraction duration_T7, Delta Onset_T7, Delta Peak-Onset Ag (s)_T7, Abs Delta Onset %Cycle_T7, Co-Contraction/ Cycle Duration_T7, Delta Peak-Onset Ant (s)_T7, EMG Rise Norm Ant (/s)_T7, Motion EMD Ant %Cycle_T7, Delta Peaks-Time Env % Cycle_T7, Delay Peaks Ant-Ag (s)_T7.

Part And Partial Correlations

Model		Partial	Part
M_1	Abs Delta Peaks-Time Env % Cycle_T7	−0.719	−0.719

Note. The intercept model is omitted, as no meaningful information can be shown.

Collinearity Diagnostics

Model	Dimension	Eigenvalue	Condition Index	Variance Proportions	
				(Intercept)	Abs Delta Peaks-Time Env % Cycle_T7
M_1	1	1.946	1.000	0.027	0.027
	2	0.054	6.028	0.973	0.973

Note. The intercept model is omitted, as no meaningful information can be shown.

Residuals Statistics

	Minimum	Maximum	Mean	SD	N
Predicted Value	−2.547	306.578	134.375	80.637	16
Residual	−126.344	188.893	1.998×10^{-15}	77.929	16
Std. Predicted Value	−1.698	2.136	1.344×10^{-16}	1.000	16
Std. Residual	−1.635	2.544	0.016	1.028	16

Squat - Linear Regression Model (w/ group) - Peak Power

Model Summary - Peak Power T2

					Durbin-Watson		
Model	R	R^2	Adjusted R^2	RMSE	Autocorrelation	Statistic	p
M_0	0.186	0.035	−0.034	114.05	−0.422	2.650	.265
M_1	0.752	0.566	0.499	79.34	−0.271	2.466	.510

ANOVA

Model		Sum of Squares	df	Mean Square	F	p
M_0	Regression	6537	1	6537	0.503	.490
	Residual	182090	14	13006		
	Total	188627	15			
M_1	Regression	106802	2	53401	8.484	.004
	Residual	81825	13	6294		
	Total	188627	15			

Coefficients

							95% CI		Collinearity Statistics	
Model		Unstandardized	Standard Error	Standardized[a]	t	p	Lower	Upper	Tolerance	VIF
M_0	(Intercept)	111.455	43.105		2.586	.022	19.004	203.907		
	Group (ABF)	40.746	57.474		0.709	.490	−82.523	164.015	1.000	1.000
M_1	(Intercept)	339.373	64.500		5.262	< .001	200.030	478.716		
	Group (ABF)	48.573	40.030		1.213	.247	−37.906	135.053	0.998	1.002
	Abs Delta Peaks-Time Env % Cycle_T7	−4.151	1.040	−0.730	−3.991	.002	−6.398	−1.904	0.998	1.002

Note. The following covariates were considered but not included: Abs Delta Onset (s)_T7, Motion EMD Ag %Cycle_T7, Co-contraction duration_T7, Delta Onset_T7, Delta Peak-Onset Ag (s)_T7, Abs Delta Onset %Cycle_T7, Co-Contraction/ Cycle Duration_T7, Delta Peak-Onset Ant (s)_T7, EMG Rise Norm Ant (/s)_T7, Motion EMD Ant %Cycle_T7, Delta Peaks-Time Env % Cycle_T7, Delay Peaks Ant-Ag (s)_T7.

[a] Standardized coefficients can only be computed for continuous predictors.

Part And Partial Correlations

Model		Partial	Part
M_0	Group	0.186	0.186
M_1	Group	0.319	0.222
	Abs Delta Peaks-Time Env % Cycle_T7	−0.742	−0.729

Collinearity Diagnostics

				Variance Proportions		
Model	Dimension	Eigenvalue	Condition Index	(Intercept)	Group (ABF)	Abs Delta Peaks-Time Env % Cycle_T7
M_0	1	1.750	1.000	0.125	0.125	
	2	0.250	2.646	0.875	0.875	
M_1	1	2.616	1.000	0.013	0.049	0.014
	2	0.332	2.807	0.032	0.925	0.057
	3	0.052	7.066	0.955	0.026	0.929

Residuals Statistics

	Minimum	Maximum	Mean	SD	N
Predicted Value	−31.940	330.432	134.375	84.381	16
Residual	−148.255	166.224	-3.553×10^{-15}	73.858	16
Std. Predicted Value	−1.971	2.323	8.544×10^{-17}	1.000	16
Std. Residual	−2.008	2.354	0.016	1.029	16

Squat - Linear Regression Model (w/ sex) - Peak Power

Model Summary - Peak Power T2

Model	R	R^2	Adjusted R^2	RMSE	Durbin-Watson		
					Autocorrelation	Statistic	p
M_0	0.000	0.000	0.000	112.14	−0.364	2.489	.310
M_1	0.719	0.517	0.483	80.66	−0.052	2.073	.918
M_2	0.839	0.704	0.659	65.51	0.115	1.708	.587

ANOVA

Model		Sum of Squares	df	Mean Square	F	p
M_1	Regression	97534	1	97534	14.99	.002
	Residual	91093	14	6507		
	Total	188627	15			
M_2	Regression	132836	2	66418	15.48	< .001
	Residual	55792	13	4292		
	Total	188627	15			

Note. The intercept model is omitted, as no meaningful information can be shown.

Coefficients

Model		Unstandardized	Standard Error	Standardized[a]	t	p	95% CI		Collinearity Statistics	
							Lower	Upper	Tolerance	VIF
M_0	(Intercept)	134.375	28.035		4.793	< .001	74.620	194.129		
M_1	(Intercept)	363.235	62.457		5.816	< .001	229.279	497.191		
	Abs Delta Peaks-Time Env % Cycle_T7	−4.089	1.056	−0.719	−3.872	.002	−6.354	−1.824	1.000	1.000
M_2	(Intercept)	360.674	50.732		7.109	< .001	251.075	470.274		
	Abs Delta Peaks-Time Env % Cycle_T7	−3.003	0.938	−0.528	−3.203	.007	−5.029	−0.978	0.837	1.195
	Gender (W)	−103.502	36.088		−2.868	.013	−181.465	−25.538	0.837	1.195

Note. The following covariates were considered but not included: Abs Delta Onset (s)_T7, Motion EMD Ag %Cycle_T7, Co-contraction duration_T7, Delta Onset_T7, Delta Peak-Onset Ag (s)_T7, Abs Delta Onset %Cycle_T7, Co-Contraction/ Cycle Duration_T7, Delta Peak-Onset Ant (s)_T7, EMG Rise Norm Ant (/s)_T7, Motion EMD Ant %Cycle_T7, Delta Peaks-Time Env % Cycle_T7, Delay Peaks Ant-Ag (s)_T7.

[a] Standardized coefficients can only be computed for continuous predictors.

Part And Partial Correlations

Model		Partial	Part
M_1	Abs Delta Peaks-Time Env % Cycle_T7	−0.719	−0.719
M_2	Abs Delta Peaks-Time Env % Cycle_T7	−0.664	−0.483
	Gender	0.623	0.433

Note. The intercept model is omitted, as no meaningful information can be shown.

Collinearity Diagnostics

Model	Dimension	Eigenvalue	Condition Index	Variance Proportions		
				(Intercept)	Abs Delta Peaks-Time Env % Cycle_T7	Gender (W)
M_1	1	1.946	1.000	0.027	0.027	
	2	0.054	6.028	0.973	0.973	
M_2	1	2.665	1.000	0.013	0.012	0.041
	2	0.285	3.057	0.08	0.028	0.884
	3	0.050	7.283	0.906	0.96	0.075

Note. The intercept model is omitted, as no meaningful information can be shown.

Residuals Statistics

	Minimum	Maximum	Mean	SD	N
Predicted Value	−11.466	319.064	134.375	94.105	16
Residual	−112.590	155.583	-1.110×10^{-15}	60.987	16
Std. Predicted Value	−1.550	1.963	7.363×10^{-17}	1.000	16
Std. Residual	−1.902	2.629	0.005	1.030	16

Squat - Linear Regression Model - Relative Peak Power (W/kg)

Model Summary - PP (W/Kg) T2

					Durbin-Watson		
Model	R	R^2	Adjusted R^2	RMSE	Autocorrelation	Statistic	p
M_0	0.000	0.000	0.000	1.425	−0.320	2.447	.354
M_1	0.681	0.464	0.425	1.080	0.014	1.946	.873

ANOVA

Model		Sum of Squares	df	Mean Square	F	p
M_1	Regression	14.12	1	14.118	12.10	.004
	Residual	16.33	14	1.167		
	Total	30.45	15			

Note. The intercept model is omitted, as no meaningful information can be shown.

Coefficients

							95% CI		Collinearity Statistics	
Model		Unstandardized	Standard Error	Standardized[a]	t	p	Lower	Upper	Tolerance	VIF
M_0	(Intercept)	1.825	0.356		5.122	< .001	1.065	2.584		
M_1	(Intercept)	4.578	0.836		5.474	< .001	2.784	6.372		
	Abs Delta Peaks-Time Env % Cycle_T7	−0.049	0.014	−0.681	−3.479	.004	−0.080	−0.019	1.000	1.000

[a] Standardized coefficients can only be computed for continuous predictors.

Part And Partial Correlations

Model		Partial	Part
M_1	Abs Delta Peaks-Time Env % Cycle_T7	−0.681	−0.681

Note. The intercept model is omitted, as no meaningful information can be shown.

Collinearity Diagnostics

				Variance Proportions	
Model	Dimension	Eigenvalue	Condition Index	(Intercept)	Abs Delta Peaks-Time Env % Cycle_T7
M_1	1	1.946	1.000	0.027	0.027
	2	0.054	6.028	0.973	0.973

Note. The intercept model is omitted, as no meaningful information can be shown.

Residuals Statistics

	Minimum	Maximum	Mean	SD	N
Predicted Value	0.177	3.896	1.825	0.970	16
Residual	−1.545	2.816	-1.908×10^{-17}	1.044	16
Std. Predicted Value	−1.698	2.136	-6.765×10^{-17}	1.000	16
Std. Residual	−1.493	2.832	0.015	1.030	16

Squat - Linear Regression Model - Relative Peak Power to Lean Mass (W/kg)

Model Summary - PP/Massa magra T2

Model	R	R^2	Adjusted R^2	RMSE	Durbin-Watson		
					Autocorrelation	Statistic	p
M_0	0.000	0.000	0.000	11.385	−0.391	2.542	.259
M_1	0.732	0.535	0.502	8.032	−0.045	2.040	.972

ANOVA

Model		Sum of Squares	df	Mean Square	F	p
M_1	Regression	1041.2	1	1041.16	16.14	.001
	Residual	903.2	14	64.51		
	Total	1944.3	15			

Note. The intercept model is omitted, as no meaningful information can be shown.

Coefficients

Model		Unstandardized	Standard Error	Standardized[a]	t	p	95% CI		Collinearity Statistics	
							Lower	Upper	Tolerance	VIF
M_0	(Intercept)	17.327	2.846		6.088	< .001	11.261	23.394		
M_1	(Intercept)	40.973	6.219		6.588	< .001	27.635	54.311		
	Abs Delta Peaks-Time Env % Cycle_T7	−0.422	0.105	−0.732	−4.017	.001	−0.648	−0.197	1.000	1.000

[a] Standardized coefficients can only be computed for continuous predictors.

Part And Partial Correlations

Model		Partial	Part
M_1	Abs Delta Peaks-Time Env % Cycle_T7	−0.732	−0.732

Note. The intercept model is omitted, as no meaningful information can be shown.

Collinearity Diagnostics

Model	Dimension	Eigenvalue	Condition Index	Variance Proportions	
				(Intercept)	Abs Delta Peaks-Time Env % Cycle_T7
M_1	1	1.946	1.000	0.027	0.027
	2	0.054	6.028	0.973	0.973

Note. The intercept model is omitted, as no meaningful information can be shown.

Residuals Statistics

	Minimum	Maximum	Mean	SD	N
Predicted Value	3.181	35.119	17.327	8.331	16
Residual	−12.472	18.617	3.157×10^{-16}	7.760	16
Std. Predicted Value	−1.698	2.136	-1.050×10^{-16}	1.000	16
Std. Residual	−1.621	2.518	0.018	1.029	16